■ Brain Tumor MRI Analysis Report v2.0

Report Generated:	2026-04-02 17:51:55
Scan File:	Te-gl_0051.jpg
Analysis Version:	2.0 (VLM Enhanced)

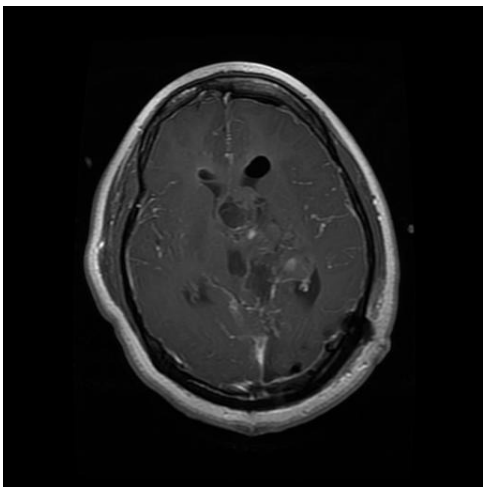
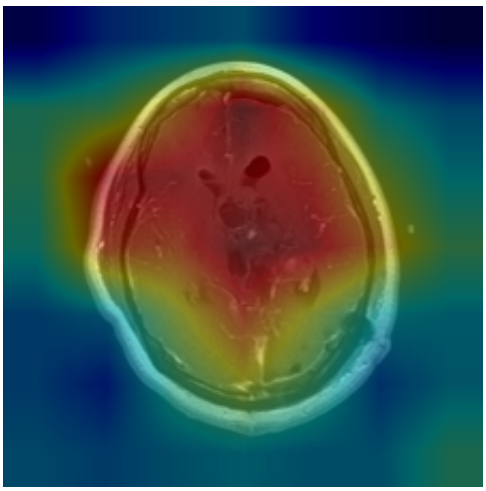
■ Classification Results

Metric	Value
Predicted Type	GLIOMA
Confidence	100.0%
Severity Level	High
Recommended Treatment	Surgery + Radiation + Chemotherapy

■ All Class Probabilities

Class	Probability
GLIOMA	100.0%
MENINGIOMA	0.0%
NOTUMOR	0.0%
PITUITARY	0.0%

■ MRI Scans & Analysis

Original MRI Scan	Grad-CAM Heatmap
	

■ AI Explanation

Source: Template-based Analysis

The classification results indicate a predicted diagnosis of glioma with 100% confidence, suggesting a high degree of certainty in the AI model's detection of this primary brain tumor. The Grad-CAM analysis reveals a focal activation pattern in the central areas, which is significant as it highlights the model's attention to the specific region of interest, consistent with the typical imaging features of glioma, such as heterogeneous enhancement, necrosis, and surrounding edema. Gliomas are often characterized by their infiltrative nature, irregular margins, and variable enhancement patterns on MRI, which are critical features that guide diagnosis and treatment planning. Given the high confidence in the classification results, it is recommended that the patient be referred to a specialist neuro-oncologist for further evaluation and management, including potential biopsy and treatment planning, to ensure timely and appropriate care.

■ Reliability & Quality Metrics

Metric	Value	Assessment
Uncertainty Score	0.000	Low
High Activation Area	21.8%	Grad-CAM focus
Top-2 Margin	100.0%	Prediction separation
Reliability Status	PASS	All checks passed

■■ Model Information

Architecture	DenseNet121

Convolutional Layer	conv5_block16_concat
Input Size	224×224 RGB
Number of Classes	4
Overall Accuracy	91.2%

■■ Medical Disclaimer

This analysis is generated by an AI system for educational and research purposes only. It should NOT be used for actual medical diagnosis or treatment decisions. Always consult qualified healthcare professionals for medical advice, diagnosis, and treatment. The AI model may produce errors or misclassifications.